

HOMEWORK 11
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2019

CHECK: 05/08/2019

SECTION 6.4 - THE GRAM-SCHMIDT PROCESS

1. Let $v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 8 \\ 5 \\ 6 \end{bmatrix}$, and write $W = \text{span}\{v_1, v_2\}$. Use the Gram-Schmidt process to find an orthogonal basis for W .
 2. Let $A = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$. Find an orthogonal basis for $\text{Col}(A)$.
-

SECTION 6.5 - LEAST-SQUARES PROBLEMS

3. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix}$. Describe all least-squares solutions to the equation $Ax = b$.
4. Let A be an $m \times n$ matrix such that $A^T A$ is invertible. Show that the columns of A are linearly independent.

(WARNING: Since A may not be a square matrix, you may **not** assume that A is invertible!)